

Full Stack Development (MERN) - Project Documentation

Project Title: AI Powered Customer Support Ticketing System

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1. Introduction

Purpose: AI-powered platform to automate customer support ticket creation, classification, prioritization and assignment.

2. Project Overview

Features: User login, ticket creation, AI categorization, priority prediction, agent dashboard, analytics, notifications.

3. Architecture

Frontend: React. Backend: Node.js + Express.js. Database: MongoDB. AI: Python NLP/ML service.

4. Setup Instructions

Prerequisites: Node.js, MongoDB, Python. Install dependencies using npm install and configure environment variables.

5. Folder Structure

Client: React UI. Server: Express APIs. AI: ML models. Database: MongoDB collections.

6. Running the Application

Frontend: npm start. Backend: npm start. AI service: python app.py.

7. API Documentation

POST /login, POST /tickets, GET /tickets, PUT /tickets/:id, POST /predict.

8. Authentication

JWT-based authentication and role-based authorization.

9. User Interface

Login, Dashboard, Ticket Form, Agent Panel, Reports.

10. Testing

Unit, integration, functional and performance testing.

11. Screenshots or Demo

Include application screenshots or demo link.

12. Known Issues

Model accuracy depends on training data quality.

13. Future Enhancements

Generative AI assistant, multilingual support, voice tickets, mobile app.